SME PROCUREMENT-PREPARATION AGENT

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A bounded agentic workflow for evidence-based reorder preparation

GROUP Y

BSE4104 Emerging Trends in Software Engineering

PROBLEM AND SCOPE

Problem context and project scope

A procurement officer must reconcile stock levels, supplier status, tax, delivery, currency and policy before preparing one reorder request. The inputs arrive in inconsistent formats and the comparison is often manual.

Effects on the workflow

01	**Delayed replenishment**	Slower response to fast-moving stock
02	**Comparison errors**	Wrong supplier economics or avoidable rework
03	**Thin audit trail**	Approvals without a defensible rationale

OBSERVED PAIN POINTS

- Stockouts surface at the counter, after demand is lost
- Supplier quotations use different units and cost assumptions
- Policy checks rely on memory, with weak justification for audit

PROJECT OBJECTIVE

System objective and engineering contribution

The system helps a procurement officer turn a weekly reorder review into a traceable requisition in DRAFT. AI handles language tasks. Deterministic code performs calculations and rule checks. People retain approval authority.

01

Detect the need

Flag stock at or below reorder point and calculate quantity deterministically.

02

Compare viable offers

Check eligibility, normalize landed cost and explain exclusions.

03

Prepare the record

Compile one auditable requisition after the officer confirms the evidence.

Core contribution **A traceable multi-step workflow that combines model-backed extraction and retrieval with bounded tools, persistent state and explicit human gates.**

WORKFLOW

End-to-end workflow and demonstration sequence

1

Scan

Cement: 150 bags on hand
120 bags required

2

Validate

Supplier eligibility and name-discrepancy checks

3

Compare

Landed cost calculated from confirmed records

4

Confirm

Officer reviews comparison and authorizes drafting

5

Draft

REQ created in DRAFT
No purchase is executed

DEMO CHECKPOINTS

Gate 2 — Comparison requires officer confirmation

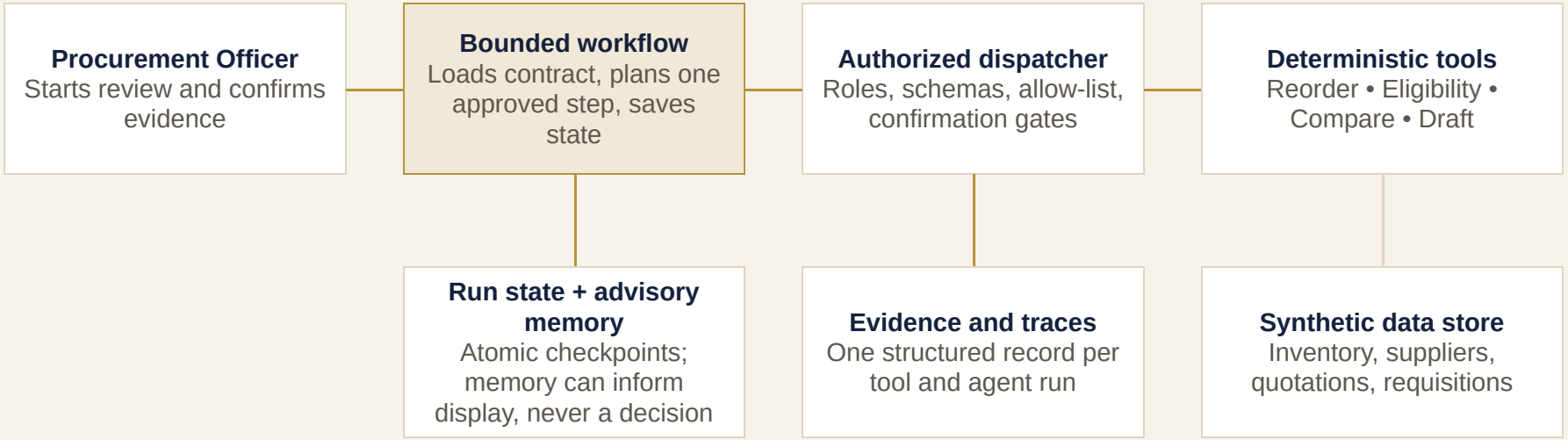
Draft confirmation — Draft write requires explicit confirmation

STOP

The workflow stops at DRAFT. There is no purchasing, supplier contact, payment or autonomous approval capability.

SYSTEM ARCHITECTURE

Implemented architecture and system boundaries



MODEL-BACKED SERVICES

Gemini extraction
11 structured fields; no arithmetic

Policy RAG
Controlled corpus; validated citations

Implementation gap

Extraction and RAG run through separate entry points. The implemented loop uses a deterministic planner.

CONTROL MODEL

AI boundary and human control

AI may

- Extract fields from converted quotation text
- Explain retrieved policy clauses
- Draft language from validated records

Code decides

- Calculate reorder quantity and landed cost
- Determine supplier eligibility and ranking
- Validate schemas, citations and every write

Humans approve

- Confirm extracted data and comparison
- Confirm the requisition draft
- Approve or reject outside the agent

Capabilities that do not exist — **Purchase order issuance • Supplier contact • Payment • Award recommendation • Programmatic approval**

PROJECT DELIVERY

Eight-week engineering progression



Boundary	Model	RAG	Tools	Agent	State	Evaluate	Harden
One workflow; synthetic data	Gemini extraction; versioned prompt	12 documents; 45 clauses	4 tools; schema contracts	Bounded loop; stop conditions	Resume, memory and fingerprints	45 scenarios; genuine failures	548 tests; limitation register

Evidence discipline

The team preserved failed runs, versioned prompt changes and re-ran the complete evaluation after fixes. Claims in this deck follow the final recorded evidence.

BOUNDED AGENCY

Agent contract, limits and recovery

PLAN
One proposal per iteration

ACT
One allow-listed tool call

OBSERVE
Validate output and scan guards

STOP / SAVE
Checkpoint, hand off or finish

HARD LIMITS

8	120 s	1	DRAFT
maximum iterations	workflow time budget	write-capable tool	highest system status

Recovery behaviour A failed step remains the current step. A safe re-run resumes from the checkpoint and does not repeat completed work or create a duplicate draft.

EVALUATION

Evaluation design and final results

45/45
scenarios passed

0
safety violations

548
offline tests passed

8/8
live calls under 30 s

Verified at release v1.0.0, 23 October 2026

chart

Hard gates held

- Zero arithmetic errors observed
- Zero requisitions beyond DRAFT
- Zero prohibited actions under adversarial prompts

Scope note: document conversion was not built. Model and RAG evidence comes from small live samples.

ENGINEERING EVIDENCE

Failure analysis and remediation

281 s

An adversarial quotation completed safely but the model call had no deadline.

REMEDIATION

30 s budget

Per-request timeout, disabled retries, watchdog and structured timeout result.

1 timeout

The clean quotation returned a clear failure at 29.95 s. The officer can rerun.

What we learned

A timeout must be bounded and visible. The 30-second deadline prevents indefinite waits, but production use still requires clear retry rules, repeated live tests and service monitoring.

READINESS

Implemented scope and known gaps

Ready now

- Synthetic-data workflow from reorder scan to DRAFT
- Deterministic calculations, eligibility and ranking
- Traceable tool runs, checkpoints and evaluation evidence
- Safe refusals, explicit hand-offs and no duplicate draft
- Optional browser operator view (no login yet)

Before real deployment

- Add authenticated roles and signed confirmations
- Use transactional, access-controlled and encrypted storage
- Integrate conversion, RAG and model planning into the loop
- Repeat live evaluation at production scale and define retry policy

Next engineering step **Integrate conversion and policy retrieval into the loop, then evaluate repeated live runs with authenticated users and production-grade storage.**

CONCLUSION

A bounded agent can support procurement preparation without taking procurement authority

Our main engineering result is the separation of model-backed language tasks from deterministic calculations, tool authorization and human approval.

LIVE DEMO

Scan
Validate
Compare
Confirm
Draft

Questions

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